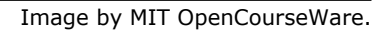


Spring 2015 – Lecture 25



REVIEW Lecture 24:

- Finite Volume on Complex geometries

- Computation of convective fluxes:

- **For mid-point rule:** $F_e = \int_{S_e} \rho \phi (\vec{v} \cdot \vec{n}) dS \approx f_e S_e = (\rho \phi \vec{v} \cdot \vec{n})_e S_e = \phi_e \dot{m}_e = \phi_e \rho_e (S_e^x u_e + S_e^y v_e)$

- Computation of diffusive fluxes: mid-point rule for complex geometries often used

- Either use shape function $\phi(x, y)$, with mid-point rule: $F_e^d \approx (k \nabla \phi \cdot \vec{n})_e S_e = k_e \left(S_e^x \frac{\partial \phi}{\partial x} \Big|_e + S_e^y \frac{\partial \phi}{\partial y} \Big|_e \right)$
- Or compute derivatives at CV centers first, then interpolate to cell faces. Option include either:

– Gauss Theorem: $\frac{\partial \phi}{\partial x_i} \bigg|_P \approx \overline{\frac{\partial \phi}{\partial x_i}} \bigg|_P = \int_{CV} \frac{\partial \phi}{\partial x_i} dV \bigg/ dV = \sum_{4 \text{ c faces}} \phi_c S_c^{x_i} \bigg/ dV$

– Deferred-correction approach: $F_e^d \approx k_e S_e \frac{\phi_E - \phi_P}{|\mathbf{r} - \mathbf{r}_P|} + k_e S_e \left[\overline{\nabla \phi|_e} \right]^{\text{old}} (\mathbf{n} - \mathbf{i}_\xi)$

– Comments on 3D

where $\left[\overline{\nabla\phi}\right]_e^{\text{old}}$ is interpolated from $\left[\overline{\nabla\phi}\right]_P^{\text{old}}$, e.g. $\left.\frac{\partial\phi}{\partial x_i}\right|_P = \sum_{4\text{ } c \text{ faces}} \phi_c S_c^{x_i} / dV$

2.29 Numerical Fluid Mechanics

Spring 2015 – Lecture 25

REVIEW Lecture 24, Cont'd:

• Turbulent Flows and their Numerical Modeling

– Properties of Turbulent Flows

- Stirring and Mixing
- Energy Cascade and Scales
- Turbulent Wavenumber Spectrum and Scales

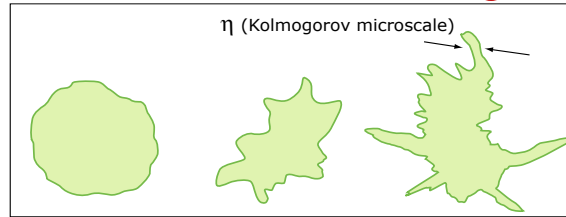


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$$L / \eta \sim O(\text{Re}_L^{3/4})$$

$$\text{Re}_L = UL/\nu$$

$$S = S(K, \varepsilon) = A \varepsilon^{2/3} K^{-5/3} \quad L^{-1} = \ell^{-1} \ll K \ll \eta^{-1}$$

ε = turbulent energy dissipation

– Numerical Methods for Turbulent Flows: Classification

- Direct Numerical Simulations (DNS) for Turbulent Flows
- Reynolds-averaged Navier-Stokes (RANS)
 - Mean and fluctuations
 - Reynolds Stresses
 - Turbulence closures: Eddy viscosity and diffusivity, Mixing-length Models, k - ε Models
 - Reynolds-Stress Equation Models
- Large-Eddy Simulations (LES)
 - Spatial filtering
 - LES subgrid-scale stresses

• Examples

References and Reading Assignments

- Chapter 9 on “Turbulent Flows” of “J. H. Ferziger and M. Peric, Computational Methods for Fluid Dynamics. Springer, NY, 3rd ed., 2002”
- Chapter 3 on “Turbulence and its Modelling” of H. Versteeg, W. Malalasekera, An Introduction to Computational Fluid Dynamics: The Finite Volume Method. Prentice Hall, Second Edition.
- Chapter 4 of “I. M. Cohen and P. K. Kundu. Fluid Mechanics. Academic Press, Fourth Edition, 2008”
- Chapter 3 on “Turbulence Models” of T. Cebeci, J. P. Shao, F. Kafyeke and E. Laurendeau, Computational Fluid Dynamics for Engineers. Springer, 2005
- Durbin, Paul A., and Gorazd Medic. Fluid dynamics with a computational perspective. Cambridge University Press, 2007.

Direct Numerical Simulations (DNS) for Turbulent Flows

- Most accurate approach
 - Solve NS with no averaging or approximation other than numerical discretizations whose errors can be estimated/controlled
- Simplest conceptually, all is resolved:
 - Size of domain must be at least a few times the distance L over which fluctuations are correlated (L = largest eddy scale)
 - Resolution must capture all kinetic energy dissipation, i.e. grid size must be smaller than viscous scale, the Kolmogorov scale, η
 - For homogenous isotropic turbulence, uniform grid is adequate, hence number of grid points (DOFs) in each direction is (Tennekes and Lumley, 1976):

$$L / \eta \sim O(\text{Re}_L^{3/4}) \quad \text{Re}_L = UL/\nu$$

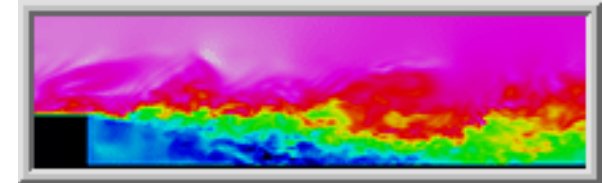
- In 3D, total cost (if time-step scales as grid size, for stability and/or accuracy):
$$\sim O\left(\text{Re}_L^{3/4} \left(\text{Re}_L^{3/4}\right)^3\right) = O(\text{Re}_L^3)$$
- CPU and RAM limit the size of the problem:
 - 10 Peta-Flops/s for 1 hour: $\text{Re}_L \sim 3.3 \cdot 10^6$ (extremely optimistic)
 - Largest DNS ever performed up to 2013: 15360 x 1536 x 11520 mesh, $\text{Re}_L = 5200$
 - <https://www.alcf.anl.gov/articles/first-mira-runs-break-new-ground-turbulence-simulations>

Direct Numerical Simulations (DNS) for Turbulent Flows: Numerics

- DNS likely gives more information that many engineers need (closer to experimental data)
- But, it can be used for turbulence studies, e.g. coherent structures dynamics and other fundamental research
 - Allow to construct better RANS models or even correlation models

- Numerical Methods for DNS

- All NS solvers we have seen are useful
- Small time-steps required for bounded errors:
 - Explicit methods are fine in simple geometries (stability satisfied due to small time-step needed for accuracy)
 - Implicit methods near boundaries or complex geometries (large derivatives in viscous terms normal to the walls can lead to numerical instabilities $\Rightarrow$ treated implicitly)



DNS, Backward facing step
Le and Moin (2008)

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Direct Numerical Simulations (DNS) for Turbulent Flows: Numerics, Cont'd

- Time-marching methods commonly used
 - Explicit 2nd to 4th order accurate (Runge-Kutta, Adams-Bashforth, Leapfrog): R-K's often more accurate for same cost
 - For same order of accuracy, R-K's allow larger time-steps
 - Crank-Nicolson often used for implicit schemes
- Must be conservative, including kinetic energy
- Spatial discretization schemes should have low dissipation
 - Upwind schemes often too diffusive: error larger than molecular diffusion!
 - High-order finite difference
 - Spectral methods (use Fourier series to estimate derivatives)
 - Mainly useful for simple (periodic) geometries (FFT)
 - Use spectral elements instead (Patera, Karniadakis, etc)
 - (Hybridizable)-(Discontinuous)-Galerkin (FE Methods): Cockburn et al.

Direct Numerical Simulations (DNS) for Turbulent Flows: Numerics, Cont'd

- Challenges:

- Storage for states at intermediate time steps ($\Rightarrow$ R-K's of low storage)
- Total discretization error and turbulence spectrum
 - Total error: both order of discretization and values of derivatives (spectrum)
 $\Rightarrow$ Measure of total error: integrate over whole turbulent spectrum
- Difficult to measure accuracy due to (unstable) nature of turbulent flow
 - Due to predictability limit of turbulence
 - Hence, statistical properties of two solutions are often compared
 - Simplest measure: turbulent spectrum
- Generating initial conditions: as much art as science
 - Initial conditions remembered over significant “eddy-turnover” time
 - Data assimilation, smoothing schemes to obtain ICs
- Generating boundary conditions
 - Periodic for simple problems, Radiating/Sponge conditions for realistic cases

Example: Spatial Decay of Turbulence Created by an Oscillating Boundary

Briggs et al (1996)

- Oscillating grid on top of quiescent fluid creates turbulence
- Decays in intensity away from grid by “turbulent diffusion”: stirring + mixing
- Used spectral method, periodic, 3rd order R-K
- DNS results agree with data
 - Used to test turbulence “closure” models
 - Did not work well because not derived for that “type” of turbulence



Fig. 9.1. Contours of the kinetic energy on a plane in the flow created by an oscillating grid in a quiescent fluid; the grid is located at the top of the figure. Energetic packets of fluid transfer energy away from the grid region. From Briggs et al. (1996)

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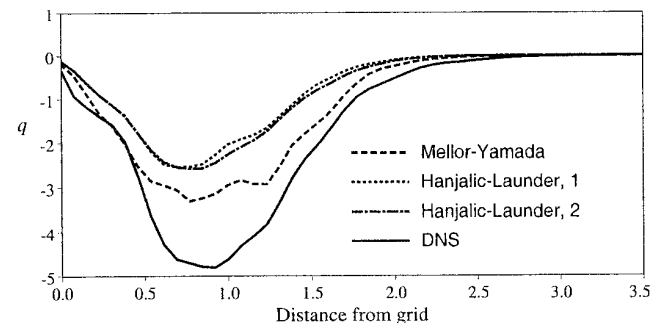


Fig. 9.2. The profile of the flux of turbulent kinetic energy, q , compared with the predictions of some commonly used turbulence models (Mellor and Yamada, 1982; Hanjalic and Launder, 1976 and 1980); from Briggs et al. (1996)

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Source: Ferziger, J. and M. Peric. *Computational Methods for Fluid Dynamics*. 3rd ed. Springer, 2001.

Note: DNS have at times found out when laboratory set-up was not proper

Reynolds-averaged Navier-Stokes (RANS)

- Many science and engineering applications focus on averages
- RANS models: based on ideas of Osborne Reynolds
 - All “unsteadiness” regarded as part of turbulence and averaged out
 - By averaging, nonlinear terms in NS eqns. lead to new product terms that must be modeled
- Separation into mean and fluctuations ($\tau \ll T_0 \ll T$)

- Moving time-average: $u = \bar{u} + u'$;

$$\phi(x_i, t) = \bar{\phi}(x_i, t) + \phi'(x_i, t)$$

where $\bar{\phi}(x_i, t) = \frac{1}{T_0} \int_{t-T_0/2}^{t+T_0/2} \phi(x_i, t) dt$; i.e. $\bar{\phi}'(x_i, t) = 0$

- Ensemble average:

$$\langle \phi(x_i, t) \rangle = \frac{1}{N} \sum_{r=1}^N \phi^r(x_i, t)$$

- Reynolds-averaging: either of the above two averages

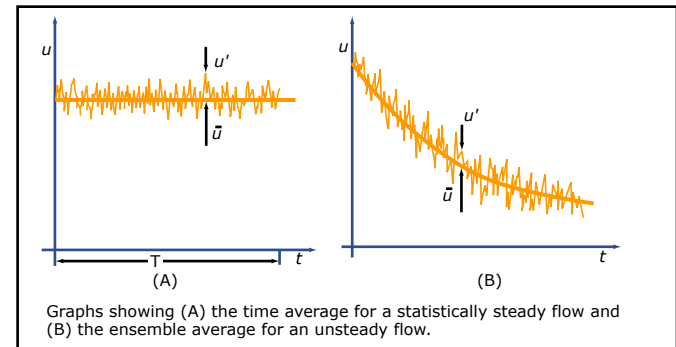


Image by MIT OpenCourseWare.

Reynolds-averaged Navier-Stokes (RANS)

- Variance, r.m.s. and higher-moments:

$$\overline{\phi'^2}(x_i, t) = \frac{1}{T_0} \int_{t-T_0/2}^{t+T_0/2} \phi'^2(x_i, t) dt \quad \overline{u_i \phi} = \overline{(\bar{u}_i + u')(\bar{\phi} + \phi')} = \bar{u}_i \bar{\phi} + \overline{u'_i \phi'}$$

$$\phi_{\text{rms}} = \sqrt{\overline{\phi'^2}}$$

$$\overline{\phi'^p}(x_i, t) = \frac{1}{T_0} \int_{t-T_0/2}^{t+T_0/2} \phi'^p(x_i, t) dt \quad \text{with } p = 3, 4, \text{ etc}$$

- Correlations:

– In time: $R(t_1, t_2; x) = \overline{u'(x, t_1) u'(x, t_2)}$ for a stationary process : $R(\tau; x) = \overline{u'(x, t) u'(x, t + \tau)}$

– In space:

$R(x_1, x_2; t) = \overline{u'(x_1, t) u'(x_2, t)}$ for a homogeneous process : $R(\ell; t) = \overline{u'(x, t) u'(x + \ell, t)}$

- Turbulent kinetic energy: $k = \frac{1}{2} \overline{u'^2 + v'^2 + w'^2}$

– Note: some arbitrariness in the decomposition $u = \bar{u} + u'$; $\phi = \bar{\phi} + \phi'$

and in the definition that “fluctuations = turbulence”

Reynolds-averaged Navier-Stokes (RANS)

- Continuity and Momentum Equations, incompressible:

$$\frac{\partial \rho u_i}{\partial x_i} = 0$$

$$\frac{\partial \rho u_i}{\partial t} = - \frac{\partial (\rho u_i u_j)}{\partial x_j} - \frac{\partial p}{\partial x_i} + \frac{\partial \tau_{ij}}{\partial x_j}$$

- Applying either the ensemble or time-averages to these equations leads to the RANS eqns.
 - In both cases, averaging any linear term in conservation equation gives the identical term, but for the average quantity
- Average the equations, inserting the decomposition: $u_i = \overline{u_i} + u'_i$
 - the time and space derivatives commute with the averaging operator

$$\overline{\frac{\partial u_i}{\partial x_i}} = \frac{\partial \overline{u_i}}{\partial x_i}$$
$$\overline{\frac{\partial u_i}{\partial t}} = \frac{\partial \overline{u_i}}{\partial t}$$

Reynolds-averaged Navier-Stokes (RANS)

- Averaged continuity and momentum equations:

$$\frac{\partial \bar{\rho u}_i}{\partial x_i} = 0$$

$$\frac{\partial \bar{\rho u}_i}{\partial t} + \frac{\partial (\bar{\rho u}_i \bar{u}_j + \overline{\rho u'_i u'_j})}{\partial x_j} = -\frac{\partial \bar{p}}{\partial x_i} + \frac{\partial \bar{\tau}_{ij}}{\partial x_j}$$

where $\bar{\tau}_{ij} = \mu \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right)$

(incompressible,
no body forces)

- For a scalar conservation equation

$$\frac{\partial \bar{\rho \phi}}{\partial t} + \frac{\partial (\bar{\rho \phi} \bar{u}_j + \overline{\rho \phi' u'_j})}{\partial x_j} = \frac{\partial}{\partial x_j} \left(\Gamma \frac{\partial \bar{\phi}}{\partial x_j} \right)$$

- e.g. for $\bar{\phi} = c_p \bar{T} \Rightarrow$ mean internal energy
- Terms that are products of fluctuations remain:
 - Reynolds stresses: $-\overline{\rho u'_i u'_j}$
 - Turbulent scalar flux: $-\overline{\rho u'_i \phi'}$
- Equations are thus not closed (more unknown variables than equations)
 - Closure requires specifying $\overline{\rho u'_i u'_j}$ and $\overline{\rho u'_i \phi'}$ in terms of the mean quantities and/or their derivatives (any Taylor series decomposition of mean quantities)

Reynolds Stresses: $\tau_{ij}^{\text{Re}} = -\rho \overline{u'_i u'_j}$

al stress acting on mean flow: $\tau_{ij} = \overline{\tau_{ij}} - \rho \overline{u'_i u'_j} = \overline{\tau_{ij}} + \tau_{ij}^{\text{Re}}$

$$\overline{\tau_{ij}} = \mu \left(\frac{\partial \overline{u_i}}{\partial x_j} + \frac{\partial \overline{u_j}}{\partial x_i} \right) \quad \text{and} \quad \tau^{\text{Re}} = -\rho \begin{bmatrix} \overline{u'^2} & \overline{u' v'} & \overline{u' w'} \\ \overline{v' u'} & \overline{v'^2} & \overline{v' w'} \\ \overline{w' u'} & \overline{w' v'} & \overline{w'^2} \end{bmatrix}$$

– If turbulent fluctuations are isotropic:

- Off diagonal elements of τ^{Re} cancel
- Diagonal elements equal: $\overline{u'^2} = \overline{v'^2} = \overline{w'^2}$

– Average of product of fluctuations not zero

- Consider mean shear flow: $\frac{\partial \overline{u}}{\partial y} > 0$
- If parcel is going up ($v' > 0$), it slows down neighbors, hence $u' < 0$ (opposite for $v' < 0$)
- Hence: $\overline{u' v'} < 0$ for $\frac{\partial \overline{u}}{\partial y} > 0$ (acts as turb. “diffus.”)

– Other meanings of Reynolds stress:

- Rate of mean momentum transfer by turb. fluctuations
- Average flux of j -momentum along i -direction

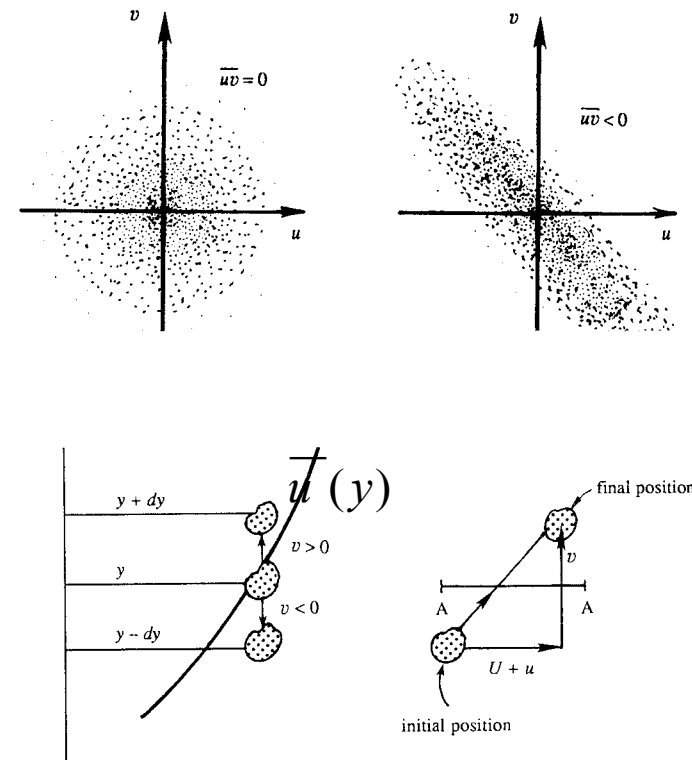


Figure 13.7 Movement of a particle in a turbulent shear flow.

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Simplest Turbulence Closure Model

y Viscosity and Eddy Diffusivity Models

- Effect of turbulence is to increase stirring/mixing on the mean-fields, hence increase effective viscosity or effective diffusivity
- Hence, “Eddy-viscosity” Model and “Eddy-diffusivity” Model

$$\tau_{ij}^{\text{Re}} = -\rho \overline{u'_i u'_j} \cong \mu_t \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right) - \frac{2}{3} \rho k \delta_{ij}$$

$$-\rho \overline{u'_i \phi'_j} \cong \Gamma_t \frac{\partial \bar{\phi}}{\partial x_j}$$

- Last term in Reynolds stress is required to ensure correct results for the sum of normal stresses:

$$\begin{aligned} \tau_{ii}^{\text{Re}} = -\rho \overline{u'_i u'_i} &= \mu_t 2 \frac{\partial \bar{u}_i}{\partial x_i} - \frac{2}{3} \rho k 3 = 0 - 2\rho \frac{1}{2} \overline{u'^2 + v'^2 + w'^2} = -\rho \left(\overline{u'^2} + \overline{v'^2} + \overline{w'^2} \right) \\ &\equiv -2\rho k \end{aligned}$$

- The use of scalar $\mu_t, \Gamma_t \Rightarrow$ assumption of isotropic turbulence, which is often inaccurate
- Since turbulent transports (momentum or scalars, e.g. internal energy) are due to “average stirring” or “eddy mixing”, we expect similar values for μ_t and Γ_t . This is the so-called Reynolds analogy, i.e. Turbulent Prandtl number ~ 1 :

$$\sigma_t = \frac{\mu_t}{\Gamma_t} \cong 1$$

Turbulence Closures: Mixing-Length Models

ing length models attempt to vary unknown μ_t as a function of position

- Main parameters available: turbulent kinetic energy k [m^2/s^2] or velocity u^* , large eddy length scale L

=> Dimensional analysis:

$$k = (u^*)^2 / 2$$

$$\mu_t = C_\mu \rho u^* L$$

C_μ dimensionless constant

- Observations and assumptions:

- Most k is contained in largest eddies of mixing-length L
- Largest eddies interact most with mean flow

$$\Rightarrow u^* = f(\bar{u}_i, \frac{\partial \bar{u}_i}{\partial x_j}, \frac{\partial^2 \bar{u}_i}{\partial x_j^2}, \dots)$$

$$\Rightarrow \text{in } \sim 2\text{D, mostly } \tau_{xy}^{\text{Re}} = -\rho \overline{u'v'} \Rightarrow$$

$$u^* \cong f\left(\frac{\partial \bar{u}}{\partial y}\right) = c L \left| \frac{\partial \bar{u}}{\partial y} \right|$$

- Hence, $\mu_t \cong \rho L^2 \left| \frac{\partial \bar{u}}{\partial y} \right|$. This is Prandtl's “mixing length” model.

- Similar to mean-free path in thermodyn: distance before parcel “mixes” with others

- For a plate flow, Prandtl assumed: $L \approx \kappa y \Rightarrow u^* \cong \kappa y \frac{\partial \bar{u}}{\partial y} \Rightarrow \frac{\bar{u}}{u^*} = \frac{1}{\kappa} \ln y + \text{const.}$

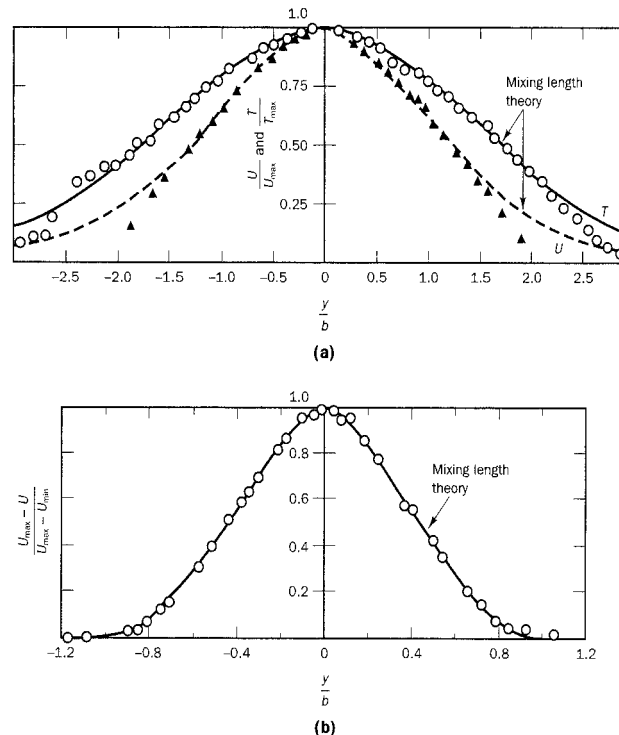
- Mixing-length turbulent Reynolds stress: $\tau_{xy}^{\text{Re}} = -\rho \overline{u'v'} \cong \rho L^2 \left| \frac{\partial \bar{u}}{\partial y} \right| \frac{\partial \bar{u}}{\partial y}$
- Mixing length model can also be used for scalars: $-\rho \overline{u'\phi'} \cong \frac{\mu_t}{\sigma_t} \frac{\partial \bar{\phi}}{\partial x} \cong \rho \frac{L^2}{\sigma_t} \left| \frac{\partial \bar{u}}{\partial y} \right| \frac{\partial \bar{\phi}}{\partial y}$

Mixing Length Models: What is L (ℓ_m)?

Simple 2D flows, mixing-length models agree well with data

- In these flows, mixing length L proportional to physical size (D , *etc*)
- Here are some examples:

Figure 3.14 Results of calculations using mixing length model for (a) planar jet and (b) wake behind a long, slender, circular cylinder
Source: Schlichting, H. (1979) *Boundary Layer Theory*, 7th edn, reproduced with permission of The McGraw-Hill Companies



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Source: Schlichting, H. *Boundary Layer Theory*. 7th ed. The McGraw-Hill Companies, 1979.

- But, in general turbulence, more than one space and time scale!

Turbulence Closures: $k - \varepsilon$ Models

- Mixing-length = “zero-equation” Model
- One might find a PDE to compute $\tau_{ij}^{\text{Re}} = -\rho \overline{u'_i u'_j}$ and $-\rho \overline{u'_i \phi'}$ as a function of k and other turbulent quantities
 - Turbulence model requires at least a length scale and a velocity scale, hence two PDEs?
- Kinetic energy equations (incompressible flows)
 - Define Total KE = $\frac{1}{2}(\overline{u_i})^2 + \frac{1}{2}(\overline{u'_i})^2$, $K = \frac{1}{2}(\overline{u_i})^2$ and $\overline{\tau_{ij}} = \mu \left(\frac{\partial \overline{u_i}}{\partial x_j} + \frac{\partial \overline{u_j}}{\partial x_i} \right) = 2\mu \overline{e_{ij}}$
 - Mean KE: Take mean mom. eqn., multiply by $\overline{u_i}$ to obtain:

$$\frac{\partial \rho K}{\partial t} + \frac{\partial (\rho K \overline{u_j})}{\partial x_j} = \frac{\partial (-\overline{p} \overline{u_j})}{\partial x_j} + \frac{\partial (2\mu \overline{e_{ij}} \overline{u_i} - \rho \overline{u'_i u'_j} \overline{u_i})}{\partial x_j} - 2\mu \overline{e_{ij}} \overline{e_{ij}} + \rho \overline{u'_i u'_j} \frac{\partial \overline{u_i}}{\partial x_j}$$

Rate of change of K	+	Advection of K	=	“Transport of K by pressure” ($\overline{p}$ work)	+	Transport of K by mean viscous stresses	+	Transport of K by Reynolds stresses	-	Rate of viscous dissipation of K	-	Rate of decay of K due turbulence production
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Turbulence Closures: $k - \varepsilon$ Models, Cont'd

kinetic energy equation

- Obtain momentum eq. for the turbulent velocity u'_i (total eq. - mean eq.)
- Define the fluctuating strain rate: $e'_{ij} \equiv \frac{1}{2} \left(\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right)$
- Multiply by u'_i (sum over i) and average to obtain the eqn. for $k = \frac{1}{2} \overline{u'_i u'_i}$

$$\frac{\partial \rho k}{\partial t} + \frac{\partial (\rho k \bar{u}_j)}{\partial x_j} = \frac{\partial (-\overline{p' u'_j})}{\partial x_j} + \frac{\partial \left(2\mu \overline{e'_{ij} u'_i} - \rho \frac{1}{2} \overline{u'_i u'_j \cdot u'_i} \right)}{\partial x_j} - 2\mu \overline{e'_{ij} \cdot e'_{ij}} - \rho \overline{u'_i u'_j} \frac{\partial \bar{u}_i}{\partial x_j}$$

Rate of change of k	+	Advection of k	=	"Transport of k by turbulent pressure"	+	Transport of k by viscous stresses (diffusion of k)	+	Transport of k by Reynolds stresses	-	Rate of viscous dissipation of k	+	Rate of production of k
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- This equation is similar than that of K , but with prime quantities
- Last term is now opposite in sign: is the rate of shear production of k : $P_k = -\rho \overline{u'_i u'_j} \frac{\partial \bar{u}_i}{\partial x_j}$
- Next to last term = rate of viscous dissipation of k : $\varepsilon = 2\nu \overline{e'_{ij} \cdot e'_{ij}}$ ($= 2\mu \overline{e'_{ij} \cdot e'_{ij}}$ p.u.m)
- These two terms often of the same order (this is how Kolmogorov microscale is defined)
 - e.g. consider steady state turbulence (steady k)
- If Boussinesq fluid, the 2 KE eqs. also contain buoyant loss/production terms

Turbulence Closures: $k - \varepsilon$ Models, Cont'd

parameterizations for the standard k equation:

- For incompressible flows, the viscous transport term is: $\frac{\partial (2\mu \overline{e'_{ij} u'_i})}{\partial x_j} = \frac{\partial}{\partial x_j} \left(\mu \frac{\partial k}{\partial x_j} \right)$
- The other two turbulent energy transport terms are thus modeled using:

$$-\overline{p' u'_j} + \left(-\rho \frac{1}{2} \overline{u'_i u'_j \cdot u'_i} \right) \approx \frac{\mu_t}{\sigma_k} \frac{\partial k}{\partial x_j} \quad (\sigma_t = \frac{\mu_t}{\Gamma_t} \cong 1)$$

- This is analogous to an “eddy-diffusion of a scalar” model, recall: $-\rho \overline{u'_i \phi'_j} \cong \Gamma_t \frac{\partial \overline{\phi}}{\partial x_j}$
- In some models, eddy-diffusions are tensors
- The production term: using again the eddy viscosity model for the Rey. Stresses

$$P_k = -\rho \overline{u'_i u'_j} \frac{\partial \overline{u}_i}{\partial x_j} \approx \left(\mu_t \left(\frac{\partial \overline{u}_i}{\partial x_j} + \frac{\partial \overline{u}_j}{\partial x_i} \right) - \frac{2}{3} \rho k \delta_{ij} \right) \frac{\partial \overline{u}_i}{\partial x_j} = \mu_t \left(\frac{\partial \overline{u}_i}{\partial x_j} + \frac{\partial \overline{u}_j}{\partial x_i} \right) \frac{\partial \overline{u}_i}{\partial x_j}$$

- All together, we have all “unknown” terms for the k equation parameterized, as long as $\varepsilon = 2\nu \overline{e'_{ij} \cdot e'_{ij}}$ the rate of viscous dissipation of k is known:

$$\frac{\partial \rho k}{\partial t} + \frac{\partial (\rho k \overline{u}_j)}{\partial x_j} = \frac{\partial}{\partial x_k} \left(\frac{\mu_t}{\sigma_t} \frac{\partial k}{\partial x_j} \right) + \frac{\partial}{\partial x_j} \left(\mu \frac{\partial k}{\partial x_j} \right) - \rho \varepsilon + \mu_t \left(\frac{\partial \overline{u}_i}{\partial x_j} + \frac{\partial \overline{u}_j}{\partial x_i} \right) \frac{\partial \overline{u}_i}{\partial x_j}$$

Turbulence Closures: $k - \varepsilon$ Models, Cont'd

- The standard $k - \varepsilon$ model equations (Launder and Spalding, 1974)

- There are several choices for $\varepsilon = 2\nu \overline{e'_{ij} \cdot e'_{ij}}$ ($[\varepsilon] = m^2 / s^3$). The standard popular one is based on the “equilibrium turbulent flows” hypothesis:

- In “equilibrium turbulent flows”, ε the rate of viscous dissipation of k is in balance with P_k the rate of production of k (i.e. the energy cascade):

$$\underline{P_k = -\rho \overline{u'_i u'_j} \frac{\partial \overline{u_i}}{\partial x_j} \approx \mu_t \left(\frac{\partial \overline{u_i}}{\partial x_j} + \frac{\partial \overline{u_j}}{\partial x_i} \right) \frac{\partial \overline{u_i}}{\partial x_j} = O\left(\mu_t (u^*)^2 / L^2\right) \approx \underline{-2\mu \overline{e'_{ij} \cdot e'_{ij}} = \rho \varepsilon}}$$

- Recall the scalings: $k = (u^*)^2 / 2$ $\mu_t = C_\mu \rho u^* L$
- This gives the length scale and the turbulent viscosity scalings:

$$\boxed{L \approx \frac{k^{3/2}}{\varepsilon} \quad \mu_t = C_\mu \rho \frac{k^2}{\varepsilon}}$$

- As a result, one can obtain an equation for ε (with a lot of assumptions):

$$\boxed{\frac{\partial \rho \varepsilon}{\partial t} + \frac{\partial (\rho \varepsilon \overline{u_j})}{\partial x_j} = \frac{\partial}{\partial x_j} \left(\frac{\mu_t}{\sigma_\varepsilon} \frac{\partial \varepsilon}{\partial x_j} \right) + C_{\varepsilon 1} P_k \frac{\varepsilon}{k} - C_{\varepsilon 2} \rho \frac{\varepsilon^2}{k}}$$

where $\mu_t = C_\mu \rho \frac{k^2}{\varepsilon}$ and $\sigma_\varepsilon, C_{\varepsilon 1}, C_{\varepsilon 2}$ are constants. The production and destruction terms of ε are assumed proportional to those of k (the ratios ε / k is for dimensions)

Turbulence Closures: $k - \varepsilon$ Models, Cont'd

- The standard $k - \varepsilon$ model (RANS) equations are thus:

$$\begin{aligned} \frac{\partial \rho k}{\partial t} + \frac{\partial(\rho k \bar{u}_j)}{\partial x_j} &= \frac{\partial}{\partial x_j} \left(\frac{\mu_t}{\sigma_k} \frac{\partial k}{\partial x_j} \right) + \frac{\partial}{\partial x_j} \left(\mu \frac{\partial k}{\partial x_j} \right) + 2\mu_t \bar{e}_{ij} \cdot \bar{e}_{ij} - \rho \varepsilon \\ \frac{\partial \rho \varepsilon}{\partial t} + \frac{\partial(\rho \varepsilon \bar{u}_j)}{\partial x_j} &= \frac{\partial}{\partial x_j} \left(\frac{\mu_t}{\sigma_\varepsilon} \frac{\partial \varepsilon}{\partial x_j} \right) + C_{\varepsilon 1} P_k \frac{\varepsilon}{k} - C_{\varepsilon 2} \rho \frac{\varepsilon^2}{k} \end{aligned}$$

with $\mu_t = C_\mu \rho \frac{k^2}{\varepsilon}$

Rate of change of k or ε + Advection of k or ε = Transport of k or ε by "eddy-diffusion" stresses + Rate of production of k or ε - Rate of viscous dissipation of k or ε

- The Reynolds stresses are obtained from: $\tau_{ij}^{\text{Re}} = -\rho \overline{u'_i u'_j} \cong \mu_t \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right) - \frac{2}{3} \rho k \delta_{ij}$
- The most commonly used values for the constants are:

$$C_\mu = 0.09, \quad C_{\varepsilon 1} = 1.44, \quad C_{\varepsilon 2} = 1.92, \quad \sigma_k = 1.0, \quad \sigma_\varepsilon = 1.3$$

- Two new PDEs are relatively simple to implement (same form as NS)
 - But, time-scales for $k - \varepsilon$ are much shorter than for the mean flow
- Other $k - \varepsilon$ models: Spalart-Allmaras $\nu - L$; Wilcox or Menter $k - \omega$; anisotropic $k - \varepsilon$'s; etc.

Turbulence Closures: $k - \varepsilon$ Models, Cont'd

merics for standard $k - \varepsilon$ models

- Since time-scales for $k - \varepsilon$ are much shorter than for the mean flow, their equations are treated separately
 - Mean-flow NS outer iteration can be first performed using old $k - \varepsilon$
 - Strongly non-linear equations for $k - \varepsilon$ are then integrated (outer-iteration) with smaller time-step and under-relaxation
- Smaller space scales requires finer-grids near walls for $k - \varepsilon$ eqns
 - Otherwise, too low resolution can lead to wiggles and negative $k - \varepsilon$
 - If grids are the same, need to use schemes that reduce oscillations
- Boundary conditions for $k - \varepsilon$ models
 - Similar than for other scalar eqns., except at solid walls
 - Inlet: k, ε given (from data or from literature)
 - Outlet or symmetry axis: normal derivatives set to zero (or other OBCs)
 - Free stream: k, ε given or zero-derivatives
 - Solid walls: depends on Re

Turbulence Closures: $k - \varepsilon$ Models, Cont'd

d-walls boundary conditions for $k - \varepsilon$ models

– No-slip BC would be standard:

- Hence, appropriate to set $k = 0$ at the wall

- But, dissipation not zero at the wall $\rightarrow$ use : $\varepsilon = \nu \frac{\partial^2 k}{\partial n^2} \Big|_{\text{wall}}$ or $\varepsilon = 2\nu \left(\frac{\partial k^{1/2}}{\partial n} \right)^2 \Big|_{\text{wall}}$

– At high-Reynolds numbers:

- One can avoid the need to solve $k - \varepsilon$ right at the wall by using an analytical shape “wall function”:

- At high-Re, in logarithmic layer :

$$L \approx \kappa y \Rightarrow u^* \cong \kappa y \frac{\partial \bar{u}}{\partial y} \Rightarrow u^+ = \frac{\bar{u}}{u^*} = \frac{1}{\kappa} \ln y + \text{const.}$$

- If dissipation balances turbulence production, recall: $L \approx \frac{k^{3/2}}{\varepsilon}$; $\mu_t = C_\mu \rho \frac{k^2}{\varepsilon}$

- Combining, one obtains: $\varepsilon \approx \frac{k^{3/2}}{L} = \frac{(u^*)^3}{\kappa y}$ and one can match: $\tau_{\text{wall}} = \rho (u^*)^2$ without resolving the viscous sub-layer

- For more details, including low-Re cases, see references

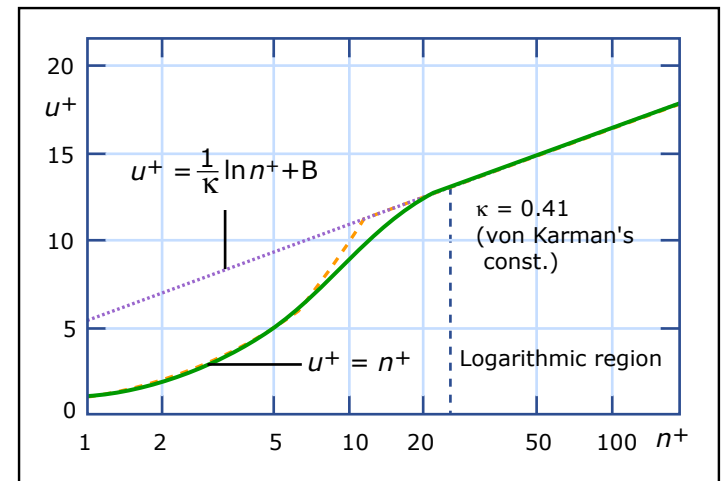


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Turbulence Closures: $k - \varepsilon$ Models, Cont'd

Example: Flow around an engine Valve (Lilek et al, 1991)

- $k - \varepsilon$ model, 2D axi-symmetric
- Boundary-fitted, structured grid
- 2nd order CDS, 3-grids refinement
- BCs: wall functions at the walls
- Physics: separation at valve throat
- Comparisons with data not bad
- Such CFD study can reduce number of experiments/tests required

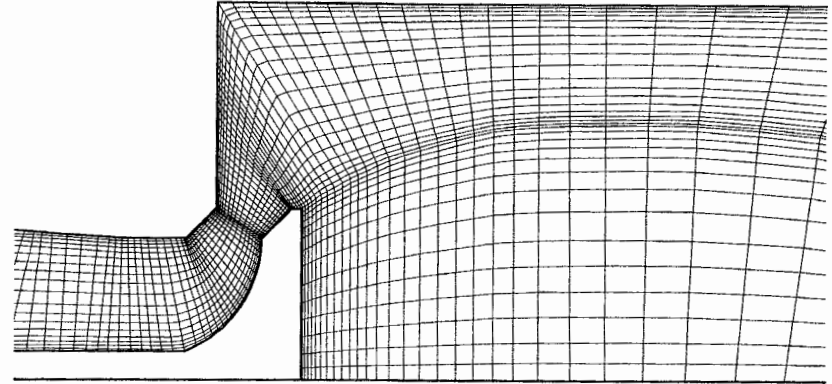


Fig. 9.12. Section of a grid (level two) used to calculate flow around a valve (from Lilek et al., 1991)

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Please also see figures 9.13 and 9.14 from Figs 9.13 and 9.14 from Ferziger, J., and M. Peric. *Computational Methods for Fluid Dynamics*. 3rd ed. Springer, 2001.

Reynolds-Stress Equation Models (RSMs)

- Underlying assumption of “Eddy viscosity/diffusivity” models and of $k - \varepsilon$ models is that of isotropic turbulence, which fails in many flows
 - Some have used anisotropic eddy-terms, but not common
- Instead, one can directly solve transport equations for the Reynolds stresses themselves: $-\rho \overline{u'_i u'_j}$ and $-\rho \overline{u'_i \phi'}$
 - These are among the most complex RANS used today. Their equations can be derived from NS
 - For momentum, the six transport eqs., one for each Reynolds stress, contain: diffusion, pressure-strain and dissipation/production terms which are unknown
 - In these “2nd order models”, assumptions are made on these terms and resulting PDEs are solved, as well as an equation for ε
 - Extra $6 + 1 = 7$ PDEs to be solved increase cost. Mostly used for academic research (assumptions on unknown terms still being compared to data)

Reynolds-Stress Equation Models (RSMs), Cont'd

Equations for $\tau_{ij}^{\text{Re}} = -\rho \overline{u'_i u'_j}$

$$\frac{\partial \tau_{ij}^{\text{Re}}}{\partial t} + \frac{\partial (\tau_{ij}^{\text{Re}} \overline{u_j})}{\partial x_j} = -\overline{p' \left(\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right)} - \left(\tau_{im}^{\text{Re}} \frac{\partial \overline{u_j}}{\partial x_m} + \tau_{jm}^{\text{Re}} \frac{\partial \overline{u_i}}{\partial x_m} \right) + \rho \varepsilon_{ij} + \frac{\partial}{\partial x_j} \left(\nu \frac{\partial \tau_{ij}^{\text{Re}}}{\partial x_j} + C_{ijm} \right)$$

Rate of change of Reynolds Stress	+	Advection of Reynolds Stress	=	Pressure-strain redistribution of Reynolds stresses	+	Rate of production of Reynolds stress	+	Rate of tensor dissipation of Reynolds stress	-	Rates of viscous and turbulent diffusions (dissipations) of Reynolds stress
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where

- The dissipation (as ε but now a tensor) is : $\varepsilon_{ij} = 2\nu \overline{e'_{ik} \cdot e'_{jk}}$
- The 3rd order turbulence diffusions are: $C_{ijm} = \rho \overline{u'_i u'_j \cdot u'_m} + \overline{p' u'_i} \delta_{jk} + \overline{p' u'_j} \delta_{ik}$
- Simplest and most common 3rd order closures:
 - Isotropic dissipation: $\varepsilon_{ij} = \frac{2}{3} \varepsilon \delta_{ij} = \frac{4}{3} \nu \overline{e'_i \cdot e'_j} \delta_{ij} \rightarrow$ one ε PDE must be solved
 - Several models for pressure-strain used (attempt to make it more isotropic), see Launder et al)
 - The 3rd order turbulence diffusions: usually modeled using an eddy-flux model, but nonlinear models also used
 - Active research

Large Eddy Simulation (LES)

- Turbulent Flows contain large range of time/space scales
- However, larger-scale motions often much more energetic than small scale ones
- Smaller scales often provide less transport

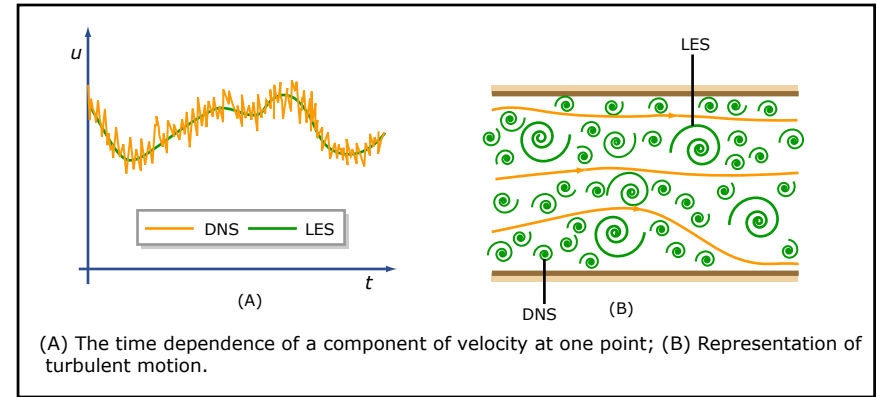


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→ simulation that treats larger eddies more accurately than smaller ones makes sense $\Rightarrow$ LES:

- Instead of time-averaging, LES uses spatial filtering to separate large and small eddies
- Models smaller eddies as a “universal behavior”
- 3D, time-dependent and expensive, but much less than DNS
- Preferred method at very high Re or very complex geometry

Large Eddy Simulation (LES), Cont'd

- Spatial Filtering of quantities

- The larger-scale (the ones to be resolved) are essentially a local spatial average of the full field

- For example, the filtered velocity is:
$$\overline{u_i}(\mathbf{x}, t) = \int_V G(\mathbf{x}, \mathbf{x}'; \Delta) u_i(\mathbf{x}', t) dV'$$

where $G(\mathbf{x}, \mathbf{x}'; \Delta)$ is the filter kernel, a localization function of support/cutoff width Δ

- Example of Filters: Gaussian, box, top-hat and spectral-cutoff (Fourier) filters

- When NS, incompressible flows, constant density is averaged, one obtains

$$\begin{aligned} \frac{\partial \overline{\rho u_i}}{\partial x_i} &= 0 \\ \frac{\partial \overline{\rho u_i}}{\partial t} + \frac{\partial (\overline{\rho u_i u_j})}{\partial x_j} &= -\frac{\partial \overline{p}}{\partial x_i} + \frac{\partial}{\partial x_j} \left[\mu \left(\frac{\partial \overline{u_i}}{\partial x_j} + \frac{\partial \overline{u_j}}{\partial x_i} \right) \right] \end{aligned}$$

- Continuity is linear, thus filtering does not change its shape

- Simplifications occur if filter does not depend on positions: $G(\mathbf{x}, \mathbf{x}'; \Delta) = G(\mathbf{x} - \mathbf{x}'; \Delta)$

Large Eddy Simulation (LES), Cont'd

sub-grid-scale stresses

- It is important to note that $\overline{\rho u_i u_j} \neq \rho \overline{u_i} \overline{u_j}$
- This quantity is hard to compute
- One introduces the sub-grid-scale Reynolds Stresses, which is the difference between the two:

$$\tau_{ij}^{SG} = -\rho \left(\overline{u_i u_j} - \overline{u_i} \overline{u_j} \right)$$

- It represents the large scale momentum flux caused by the action of the small or unresolved scales (SG is somewhat a misnomer)
- Example of models:
 - Smagorinsky: it is an eddy viscosity model

$$\tau_{ij}^{SG} - \frac{1}{3} \tau_{kk}^{SG} \delta_{ij} \cong \mu_t \left(\frac{\partial \overline{u_i}}{\partial x_j} + \frac{\partial \overline{u_j}}{\partial x_i} \right) = 2\mu_t \overline{e_{ij}}$$

- Higher-order SGS models
- More advanced models: mixed models, dynamic models, deconvolution models, etc.
- Mixed eqns, e.g. Partially-averaged Navier-Stokes (PANS): RANS $\rightarrow$ LES

Examples (see Durbin and Medic, 2009)

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2.29 Numerical Fluid Mechanics

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